

Benchmarking Forecast Rankings for Deployment-Aware Model Selection

Abstract

Forecasting benchmarks typically rank models by forecast-side metrics. In deployment, however, forecasts are often translated into actions through fixed operational interfaces, and the model preferred by forecast-side evaluation need not be the model that performs best after execution. We study this as a forecasting benchmark question: under a fixed frictional interface, does forecast-side ranking remain reliable for deployed model selection? We focus on Q2, the fixed-interface selection problem, and use Q1 only as mechanism support for why divergence can arise. Across a synthetic zero-friction anchor, a forecasting-native event-micro benchmark, a complementary Traffic-Hourly alert family, and inventory corroboration, we find that forecast-side winners can become recurrently deployed-suboptimal as friction increases. These results do not invalidate forecast-side evaluation for prediction; rather, they suggest that forecasting benchmarks aimed at deployment may need to report deployed-selection robustness alongside forecast-side metrics.

1 Introduction

Forecasting benchmarks typically compare models using forecast-side metrics. This is appropriate when the goal is to measure predictive quality. But many forecasting systems are not used directly: they are deployed through fixed interfaces that translate forecasts into actions under operational frictions such as switching costs, inertia, or execution constraints. In such settings, the model selected by forecast-side evaluation need not be the model that performs best after execution.

This paper studies that mismatch as a forecasting evaluation problem. Our main question is not how to learn a better policy, but whether forecast-side model ranking remains reliable for deployment-aware model selection under one fixed forecast-to-decision interface. We call this Q2. We separate Q2 from a mechanism question, Q1, which asks whether the same proposed action path can produce different realized outcomes under different interfaces. This separation keeps forecast-side metrics in their role as prediction measures while isolating when the object of model selection changes under deployment friction.

Forecasting benchmark work often evaluates forecast quality, calibration, and live predictive performance. Our question is complementary: whether forecast-side ranking remains trustworthy for deployed model selection under a fixed frictional interface. Unlike decision-aware forecasting or control work, we do not propose new training objectives or policies; we study the evaluation object used for model selection after forecasts are mediated by actions and friction.

This paper makes three narrow contributions.

- First, we introduce deployed-selection robustness as a forecasting benchmark question: whether forecast-side model ranking remains reliable after forecasts are executed through a fixed interface.
- Second, we document a Q2 failure mode: the forecast-side winner becomes recurrently deployed-suboptimal as friction increases under the studied fixed interfaces.
- Third, we provide a Q2-first evidence hierarchy spanning synthetic, forecasting-native event and traffic evidence, and operational corroboration; Q1 is retained only as mechanism support.

2 Two Evaluation Questions

We distinguish three evaluation objects: forecast-side quality, target-side decision quality, and realized-action quality. Forecast-side quality measures predictive performance, target-side decision quality scores the proposed action path before frictional execution, and realized-action quality scores the executed path after the fixed interface is applied. Q1 holds the proposed action path fixed and varies only the interface, so it is an exact-control mechanism check. Q2 holds the interface fixed and varies the forecaster, so it is the benchmark-selection question. Because Q2 asks whether the forecast-side winner remains the deployed winner, Q2 is the paper’s main empirical claim.

3 Evidence Design

Our evidence hierarchy is Q2-first. Synthetic provides the exact zero-friction sanity anchor. A minimal event-micro benchmark evaluated over 100 seeds provides the main forecasting-native Q2 evidence: models emit event probabilities, a fixed thresholding rule maps probabilities to actions, and switching friction penalizes action changes. A complementary Traffic-Hourly redesign family provides a second forecasting-native Q2 setting under the same fixed-interface question. In its main fixed-budget alert variant, the forecast-side winner and deployed winner coincide at zero friction but separate once switching costs are introduced, allowing the paper to test deployed-selection robustness beyond the minimal event-micro family. A relative-rank Traffic variant is retained in the appendix as forecasting-native support under the same source and pooled-model protocol. Inventory provides operational corroboration, while load-following is retained only as appendix-only secondary corroboration because its zero row remains mixed. Q1 appears only as mechanism support from synthetic and inventory and is summarized briefly in the main text.

In inventory Q2, the main text retains only the moderate-to-high friction rows from a fixed five-family comparison under one responsive replenishment rule, while the appendix records the mixed zero/low-friction rows and the broader family sweep

under the same held-out protocol.

4 Results

4.1 Q2: Fixed interface, benchmark failure and selection consequence

Q2 asks the paper’s headline benchmark question: under one fixed forecast-to-decision interface, can forecast-side ranking still be trusted for deployed model selection? Synthetic provides the exact zero-friction sanity anchor; Figure 2 focuses on the fixed-interface winner-inversion evidence in the forecasting-native and corroborative domains.

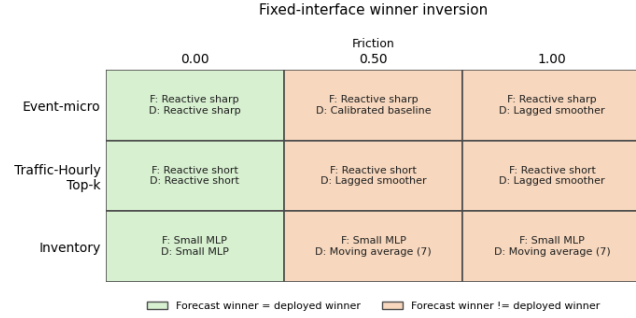


Figure 2: Forecast-side winners need not remain deployed winners. Across fixed-interface forecasting tasks, forecast-side and deployed model selection agree at zero friction in the forecasting-native evidence, but separate as switching friction increases. The benchmark consequence is deployed-suboptimal selection: the forecast-selected model is recurrently not the deployed-best model.

The event-micro table provides the main forecasting-native evidence. In a minimal event-micro benchmark evaluated over 100 seeds, agreement is highest at zero friction (0.62), falls to 0.31 by 0.50, and reaches 0.01 at 1.00. Over the same rows, the deployed winner shifts from Reactive sharp to Calibrated baseline and then Lagged smoother, while deployed-suboptimality reaches 69/100 seeds at 0.50 and 99/100 at 1.00. The event-micro pattern persists under an alternate threshold, under a hysteresis-threshold interface, and when forecast-side ranking is recomputed with log loss.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive sharp	Reactive sharp	0.62	0.003	38/100
0.50	Reactive sharp	Calibrated baseline	0.31	0.011	69/100
1.00	Reactive sharp	Lagged smoother	0.01	0.057	99/100

Table 1: Forecast-side vs. deployed-side model selection in the forecasting-native event-micro benchmark. Under a fixed thresholding interface with switching friction, the forecast-selected model becomes recurrently deployed-suboptimal as friction rises.

A complementary Traffic-Hourly fixed-budget alert family shows the same benchmark consequence under a different forecasting-native task structure. In the selected Top-k setting ($k = 249$), Reactive short is both the forecast-side winner and the deployed winner at zero friction, with exact agreement 1.0. Once switching costs are introduced, however, the forecast-side winner remains Reactive short while the deployed winner shifts to Lagged smoother, and deployed-suboptimality

reaches 100/100 seeds at both friction 0.5 and friction 1.0. A relative-rank Traffic variant shows the same qualitative pattern in the appendix.

Inventory provides operational corroboration with stronger recurrence magnitudes. Inventory has a mixed zero/low-friction regime, so we use it only as operational corroboration for the moderate-to-high friction rows. By friction 0.5, the forecast-selected model is deployed-suboptimal in 9/10 seeds and agreement falls to 0.10; at friction 1.0, deployed suboptimality rises to 10/10 seeds with mean and median deployed gaps of 1.187 and 1.162. Thus the practical issue is not merely that rankings drift, but that model choice under a fixed deployed rule becomes repeatedly wrong at moderate-to-high friction.

Domain	Friction	Forecast-side winner	Deployed winner	Deployed-suboptimal seeds
Traffic Top-k	0.00	Reactive short	Reactive short	0/100
Traffic Top-k	0.50	Reactive short	Lagged smoother	100/100
Traffic Top-k	1.00	Reactive short	Lagged smoother	100/100
Inventory	0.50	Small MLP	Moving average (7)	9/10
Inventory	1.00	Small MLP	Moving average (7)	10/10

Table 2: Compact corroboration across a complementary Traffic-Hourly alert family and the moderate-to-high friction inventory rows.

An appendix-only load-following check remains directionally consistent but retains a mixed zero row, so we do not use it as headline corroboration in the main text (Appendix G).

Appendix Tables A3–A4 and Figures A2–A3 support the interpretation that the main moderate/high-friction patterns, especially in event-micro and inventory, are not driven solely by isolated seed noise.

Appendix Tables A15–A17 widen the inventory comparison set under a standardized expanded comparison. That broader-family sweep strengthens moderate/high-friction mismatch, but the zero- and low-friction rows remain too mixed for main-text use and therefore do not overturn the narrower five-family claim.

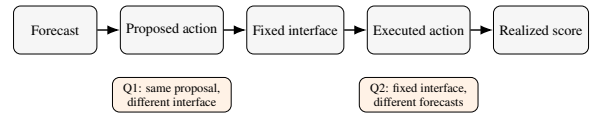


Figure 1: Forecast-to-decision evaluation setup. Forecasts are converted into proposed actions and then executed through a fixed frictional interface. Q2, the paper’s main question, asks whether forecast-side model ranking matches deployed model ranking under one fixed interface. Q1 is used only as mechanism support.

Q1 mechanism support. Q1 is mechanism support rather than the paper’s claim. Holding the proposed action path fixed, synthetic and inventory show that frictional interfaces can separate proposed and realized actions: synthetic yields an increasing proposal-versus-realization gap away from zero friction, while inventory shows a threshold pattern in which sufficiently high friction favors tempered execution. This explains why forecast-side, target-side, and realized-action evaluation can diverge; Q2 shows when that divergence becomes deployed misselection.

5 Discussion, Related Work, and Limitations

The paper’s practical claim is narrow: forecast-side metrics may remain valid for prediction, yet under a fixed deployed interface they may be insufficient for deployed model selection. Event-micro provides the main forecasting-native Q2 evidence, Traffic-Hourly provides a complementary forecasting-native family under the same fixed-interface question, inventory provides operational corroboration, and Q1 explains why the evaluation object can change under friction.

Forecasting benchmarks and leaderboards typically emphasize forecast-side accuracy, calibration, or live question resolution rather than fixed-interface deployed model selection [1, 2, 3]. Recent efforts such as ForecastBench, Prophet Arena, and GIFT-Eval broaden forecasting evaluation through dynamic, live, or zero-shot benchmark design [1, 2, 3]. Our question is complementary: under one fixed operational interface, does the forecast-side winner remain the deployed-best model? Our contribution is evaluative rather than algorithmic: we do not learn a policy or modify the training objective. We contrast with decision-focused learning and control-oriented formulations only to clarify scope: we do not optimize a policy or modify training objectives. Unlike task-based and smart predict-then-optimize methods that change training to improve downstream decisions [4, 5], we keep training fixed and study the evaluation object used for model selection.

Three limitations bound the claim. Forecasting-native Q2 evidence now spans more than one task family, though it is still not a broad benchmark suite. Inventory remains corroborative because its zero/low-friction regime is mixed, while load-following remains appendix-only secondary corroboration for the same reason. In the expanded-family inventory sweep, moderate/high-friction mismatch strengthens but zero- and low-friction rows remain mixed, so we do not use that broader comparison as main-text confirmation.

The resulting implication is evaluative rather than predictive: forecasting benchmarks aimed at deployment may benefit from reporting deployed-selection robustness alongside forecast-side metrics.

References

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A Evidence Hierarchy

Domain	Question	Fixed	Varying	Zero-fric.	Positive-fric.	Role
Synthetic	Q2	interface	forecaster	aligned at zero	ranking flips	zero-fric. anchor
Event micro	Q2	interface	forecaster	aligned at zero	selection failure recurs	main fct.-native Q2 evid.
Traffic Top-k Alert	Q2	interface	forecaster	aligned at zero	switching-cost divergence recurs	comp. second fct.-native family
Traffic Relative-Rank	Q2	interface	forecaster	aligned at zero	qualitative separation recurs	appx.-only fct.-native support
Inventory	Q2	interface	forecaster	mixed at zero	stronger recurrence directional	main oper. corrob.
Load-following	Q2	interface	forecaster	mixed at zero	recurrence	appx.-only sec. corrob.
Synthetic	Q1	proposal	interface	exact agreement	mismatch emerges	mech. support
Inventory	Q1	proposal	interface	coincide at zero	threshold story	mech. support

Table A1: Evidence map for the paper’s Q2-first hierarchy, covering the synthetic, event-micro, traffic, inventory, and load-following evidence blocks used in the paper and appendix.

B Q1 Mechanism Support

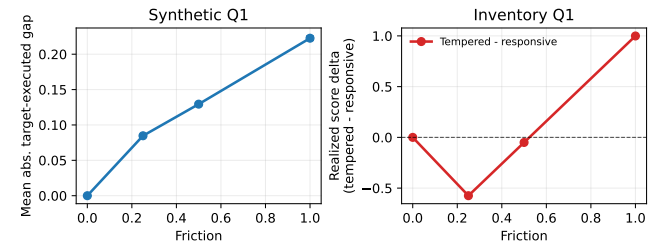


Figure A1: Appendix Q1 mechanism support. Synthetic yields exact zero-friction coincidence and a rising positive-friction proposal-versus-realization gap, while inventory shows a threshold pattern in which sufficiently high friction favors tempered execution. The figure explains why Q2 divergence can arise but is not itself the paper’s main evidence.

C Additional Q2 Reliability Checks

This appendix collects the cross-domain seed-level reliability checks and the expanded same-interface inventory comparison. These appendix-only additions do not overturn the narrower main claim from the five-family inventory comparison summarized in the main text.

The same Q2 package also admits simple seed-level reliability checks. Table A3 reports one-sided exact binomial tests for whether the forecast-selected model is deployed-suboptimal in a majority of seeds in the main recurrence rows. Table A4 reports 95% bootstrap intervals for the deployed-gap rows. Figure A2 adds uncertainty bands for the synthetic and event-micro Q2 curves, while Figure A3 shows seed-level disagreement strips for the main Q2 domains.

Domain	Friction	Deployed-suboptimal seeds / total	Share	95% exact CI	One-sided binomial p
Event micro	0.50	69/100	0.69	[0.605, 1.000]	<0.001
Event micro	1.00	99/100	0.99	[0.953, 1.000]	<0.001
Inventory	0.50	9/10	0.90	[0.606, 1.000]	0.011
Inventory	1.00	10/10	1.00	[0.741, 1.000]	<0.001
Load-following	0.25	7/10	0.70	[0.393, 1.000]	0.172
Load-following	0.50	7/10	0.70	[0.393, 1.000]	0.172
Load-following	1.00	8/10	0.80	[0.493, 1.000]	0.055

Table A3: One-sided exact binomial tests for whether the forecast-selected model is deployed-suboptimal in a majority of seeds. The strongest event-micro and inventory rows show clear majority recurrence, while load-following shows the same directional pattern with smaller seed counts.

Domain	Role	Status	Seeds	Zero flip	Zero rho	Best +fric.	Best +flip	Strongest pair	Flip share	Note
Synthetic	required	clears stricter gate	20	0.000	1.000	0.2500	0.500	Linear AR / Reactive heuristic	1.000	passed stricter gate
Inventory	required	mixed zero row	10	0.120	0.840	1.0000	0.500	Small MLP / Moving average (7)	1.000	does not clear stricter gate

Table A2: Expanded same-interface inventory comparison summary retained for appendix transparency. These statuses do not overturn the narrower main-text claim from the five-family inventory comparison.

Domain	Friction	Mean deployed gap	Mean gap 95% bootstrap CI	Median deployed gap	Median gap 95% bootstrap CI
Event micro	0.50	0.011	[0.009, 0.014]	0.008	[0.004, 0.011]
Event micro	1.00	0.057	[0.052, 0.063]	0.053	[0.048, 0.059]
Inventory	0.50	0.308	[0.186, 0.443]	0.227	[0.176, 0.484]
Inventory	1.00	1.187	[0.995, 1.390]	1.162	[0.969, 1.389]
Load-following	0.25	0.001	[0.000, 0.001]	0.000	[0.000, 0.001]
Load-following	0.50	0.001	[0.000, 0.002]	0.001	[0.000, 0.002]
Load-following	1.00	0.002	[0.001, 0.005]	0.002	[0.000, 0.003]

Table A4: 95% bootstrap intervals for deployed-gap rows in the main Q2 evidence blocks. Event-micro and inventory provide the strongest positive-gap support, while load-following remains directional secondary corroboration.

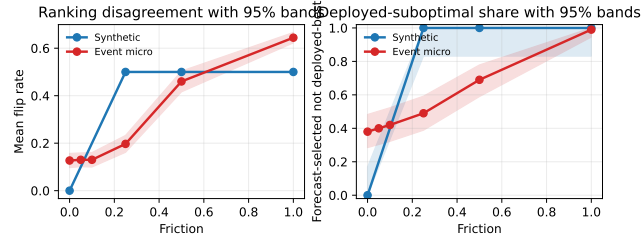


Figure A2: Appendix uncertainty bands for the synthetic and event-micro Q2 curves. Left: mean flip rate with 95% bands from seed-level standard errors. Right: deployed-suboptimal share with exact 95% binomial intervals.

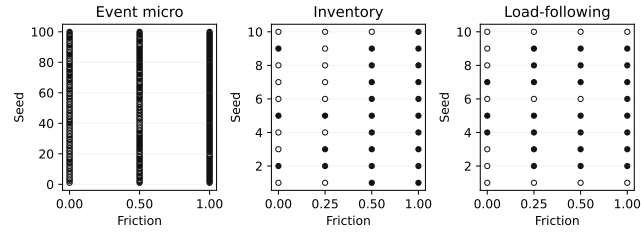


Figure A3: Seed-level disagreement strips for the main Q2 domains. Filled markers indicate seeds where the forecast-selected model is not deployed-best under the fixed interface.

D Event-Micro Robustness Package

Because event-micro provides the paper’s main forecasting-native Q2 evidence, this appendix records the first domain-specific robustness package for that benchmark. We construct a minimal forecasting-native binary-event setting evaluated over 100 seeds, in which each forecaster emits event probabilities, a fixed thresholding rule maps probabilities to actions, and switching friction penalizes action changes. The main text uses Brier as the canonical forecast-side criterion. This appendix records the full six-row Brier table plus an alternate-threshold check, a log-loss reranking check, and a second-interface hysteresis check.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap
0.00	Reactive sharp	Reactive sharp	0.62	0.003
0.05	Reactive sharp	Reactive sharp	0.60	0.002
0.10	Reactive sharp	Reactive sharp	0.58	0.002
0.25	Reactive sharp	Calibrated baseline	0.51	0.003
0.50	Reactive sharp	Calibrated baseline	0.31	0.011
1.00	Reactive sharp	Lagged smoother	0.01	0.057

Table A5: Forecast-side vs. deployed-side model selection in the full six-row event-micro benchmark summary. Under the canonical fixed-threshold interface, deployed-suboptimal counts rise from 38/100 seeds at zero friction to 99/100 at friction 1.00.

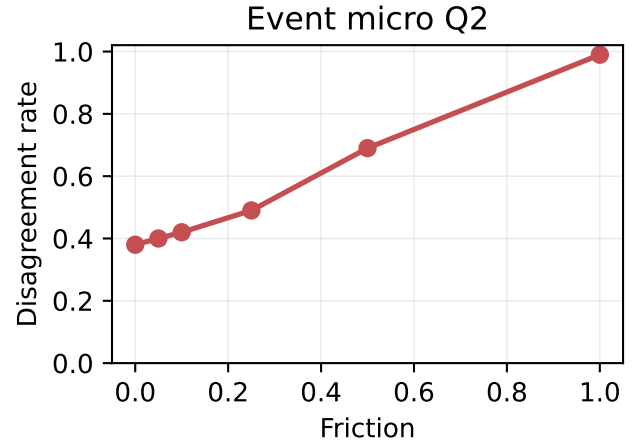


Figure A4: Ranking mismatch in a minimal event-forecasting micro-benchmark. Disagreement between forecast-side and deployed-side model selection is lowest at zero and very low friction and increases as switching frictions rise.

Threshold	Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
$\tau=0.55$	0.00	Reactive sharp	Reactive sharp	0.62	0.003	0.000	38/100
$\tau=0.55$	0.50	Reactive sharp	Calibrated baseline	0.31	0.011	0.008	69/100
$\tau=0.55$	1.00	Reactive sharp	Lagged smoother	0.01	0.057	0.053	99/100
$\tau=0.50$	0.00	Reactive sharp	Reactive sharp	0.56	0.002	0.000	44/100
$\tau=0.50$	0.50	Reactive sharp	Calibrated baseline	0.23	0.014	0.011	77/100
$\tau=0.50$	1.00	Reactive sharp	Lagged smoother	0.02	0.061	0.059	98/100

Table A6: Alternate-threshold robustness for the event-micro benchmark. The same qualitative winner drift and recurrent deployed-suboptimality pattern persists when the fixed threshold is changed from $\tau = 0.55$ to $\tau = 0.50$.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive sharp	Reactive sharp	0.62	0.002	0.000	38/100
0.50	Reactive sharp	Calibrated baseline	0.26	0.013	0.010	74/100
1.00	Reactive sharp	Lagged smoother	0.01	0.060	0.057	99/100

Table A7: Log-loss robustness for the event-micro benchmark at the canonical threshold. The same qualitative winner drift and recurrent deployed-suboptimality pattern persists when forecast-side ranking is recomputed with log loss instead of Brier.

Interface	Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
Hysteresis threshold	0.00	Reactive sharp	Reactive sharp	0.55	0.003	0.000	45/100
Hysteresis threshold	0.50	Reactive sharp	Calibrated baseline	0.45	0.005	0.001	55/100
Hysteresis threshold	1.00	Reactive sharp	Lagged smoother	0.18	0.022	0.020	82/100

Table A8: Second-interface robustness for the event-micro benchmark. Under a fixed hysteresis-threshold interface with $\delta = 0.05$, the same qualitative winner drift and deployed-suboptimality pattern remains visible, though the magnitudes are smaller than under the canonical fixed threshold.

Across this appendix robustness package, the same qualitative winner-drift and recurrent deployed-suboptimality pattern persists in a minimal forecasting-native event-micro benchmark evaluated over 100 seeds.

D.1 Interface-Aware Event-Micro Stats Addendum

Because the hysteresis-threshold check is retained paper-facing as a second-interface robustness result, this addendum records interface-specific recurrence and gap summaries without changing the main cross-domain recurrence package.

Interface	Friction	Deployed-suboptimal seeds / total	Share	95% exact CI	One-sided binomial p
Fixed threshold	0.50	69/100	0.69	[0.605, 1.000]	<0.001
Fixed threshold	1.00	99/100	0.99	[0.953, 1.000]	<0.001
Hysteresis threshold	0.50	55/100	0.55	[0.463, 1.000]	0.184
Hysteresis threshold	1.00	82/100	0.82	[0.745, 1.000]	<0.001

Table A9: Interface-aware recurrence tests for event-micro deployed-suboptimality. The canonical fixed threshold shows stronger recurrence, while hysteresis preserves the same direction with smaller magnitudes.

Interface	Friction	Mean deployed gap	Mean gap 95% bootstrap CI	Median deployed gap	Median gap 95% bootstrap CI
Fixed threshold	0.50	0.011	[0.009, 0.014]	0.008	[0.004, 0.011]
Fixed threshold	1.00	0.057	[0.052, 0.063]	0.053	[0.048, 0.059]
Hysteresis threshold	0.50	0.005	[0.004, 0.006]	0.001	[0.000, 0.003]
Hysteresis threshold	1.00	0.022	[0.019, 0.026]	0.020	[0.015, 0.025]

Table A10: Interface-aware bootstrap intervals for the event-micro deployed gap. Positive intervals remain visible under both interfaces, with smaller gaps under hysteresis.

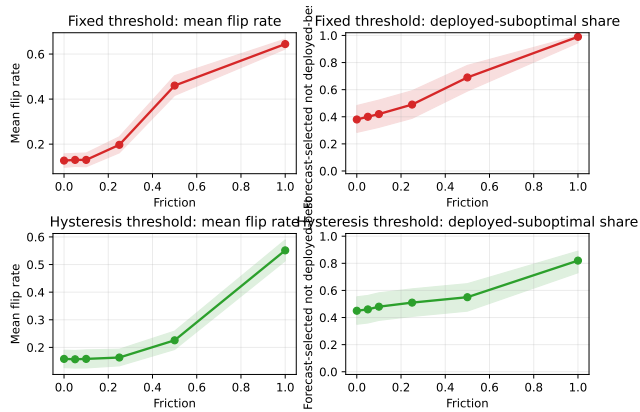


Figure A5: Interface-aware event-micro uncertainty addendum. Rows facet the canonical fixed threshold and the hysteresis threshold; columns show mean flip rate bands and deployed-suboptimal share bands.

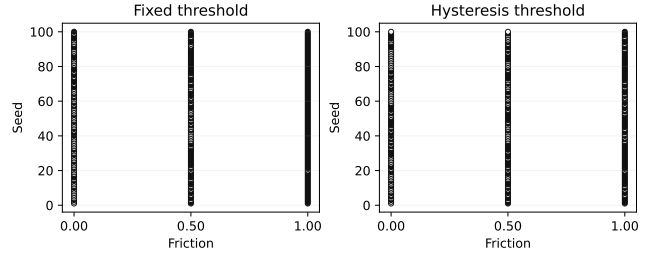


Figure A6: Interface-aware seed-level disagreement strips for the event-micro benchmark. Each panel shows the same seed set under one fixed interface.

E Traffic-Hourly Forecasting-Native Redesign Families

E.1 Fixed-Budget Top-k Alert

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive short	Reactive short	1.00	0.000	0.000	0/100
0.50	Reactive short	Lagged smoother	0.00	7.016	6.977	100/100
1.00	Reactive short	Lagged smoother	0.00	24.651	24.473	100/100

Table A11: Forecast-side vs. deployed-side model selection in a complementary fixed-budget Traffic-Hourly alert family. The forecast-side winner is deployment-optimal at zero friction, but switching costs shift the deployed winner to a smoother family.

In the selected Top-k setting, Reactive short is the forecast-side winner and the deployed winner at zero friction. At frictions 0.5 and 1.0, the deployed winner shifts to Lagged smoother even though Reactive short remains Brier-best and continues to win on the forecast side. The qualitative reading is the same as in the main text: under a fixed interface, the more reactive forecast-optimal family need not remain deployment-optimal once switching is penalized.

E.2 Relative-Rank Traffic Variant

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive short	Reactive short	1.00	0.000	0.000	0/100
0.50	Reactive short	Calibrated baseline	0.00	10.103	10.122	100/100
1.00	Reactive short	Calibrated baseline	0.00	25.205	25.231	100/100

Table A12: Relative-rank Traffic redesign retained as forecasting-native appendix support. The same qualitative separation between forecast-side and deployed-side selection appears under fixed-budget set action.

The relative-rank target shows the same qualitative separation under a different forecasting-native label construction. Reactive short remains the forecast-side winner, while Calibrated baseline becomes the deployed winner once switching costs are introduced, so we retain this variant as appendix-only forecasting-native support. The alternate $m = 0.15$ variant is also strong in the full report but is omitted from the compact table to keep the appendix narrow.

F Inventory Operational Corroboration

This appendix records the full five-family inventory summary that underlies the compact corroboration rows in the main text. The mixed zero and low-friction rows are retained here for completeness, while the main text uses only the moderate-to-high friction rows to avoid overstating the zero-friction story.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Small MLP	Small MLP	0.70	0.022	3/10
0.25	Small MLP	Linear AR	0.70	0.053	3/10
0.50	Small MLP	Moving average (7)	0.10	0.308	9/10
1.00	Small MLP	Moving average (7)	0.00	1.187	10/10

Table A13: Full inventory Q2 summary retained as appendix corroboration under a fixed replenishment interface.

G Load-Following as a Second Operational Corroboration Domain

We retain a second operational forecasting domain based on aggregated electricity load-following only as appendix-only secondary corroboration. Forecasts are translated into operational actions under a fixed responsive interface, so this section follows the event-micro robustness package as a diagnostic appendix support block rather than as additional main evidence. Using UCI ElectricityLoadDiagrams20112014, we build an aggregate multi-client load-following domain by summing disjoint client groups and evaluating at the selected 60-minute operational resolution. Groups 0 and 1 are used only for restricted calibration, while groups 2–9 serve as held-out evaluation units, so this appendix replaces the earlier household proxy with a stronger cross-sectional operational design.

A grouping-only balance-repair rerun substantially reduces group imbalance and Q1 target clipping while preserving the Q2 drift structure. A final restricted cap/margin search, however, finds no jointly feasible configuration that also satisfies the remaining Q1 requirement, so the domain remains appendix-only secondary corroboration rather than additional Q1 promotion evidence.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Linear AR	Small MLP	0.70	0.000	0.000	3/10
0.25	Linear AR	Small MLP	0.30	0.001	0.000	7/10
0.50	Linear AR	Small MLP	0.30	0.001	0.001	7/10
1.00	Linear AR	Small MLP	0.20	0.002	0.002	8/10

Table A14: Load-following Q2 summary retained as appendix-only secondary corroboration under a fixed responsive interface.

This mixed zero row is not a tie-policy artifact: the zero-row winning sets are singletons in all seeds, yet the selected candidate still shows 3/10 zero-friction mismatches with mean deployed gap 0.0002 and median gap 0.0. The grouping-only balance repair lowers the zero-row mean gap to about 0.0001 but does not remove the mismatch, so the most plausible reading is near-zero seed instability or grouping granularity rather than a clean zero-friction story.

From friction 0.25 onward, the forecast-side winner remains Linear AR while the deployed winner is Small MLP, agreement stays at 0.30, 0.30, and 0.20, and deployed-suboptimal counts rise to 7/10, 7/10, and 8/10. The domain therefore contributes directionally consistent secondary

corroboration only; it does not serve as main evidence or as additional Q1 promotion evidence in the current round.

H Stronger Inventory Q2 Baselines

To reduce baseline-specification ambiguity, we widen the inventory Q2 comparison set with stronger classical and neural families while leaving the environment, split, friction grid, fixed responsive interface, deployed metric, and tie policy unchanged. This appendix section is included as a broader transparency layer rather than as earlier or stronger main evidence than the five-family inventory comparison. All tunable families share the same held-out protocol, validation criterion, and standardized two-stage tuning structure, while the two simple heuristics remain fixed ex ante. Family representatives are selected by validation negative MAE only, and the held-out forecast-side winners in the robustness table are then computed from the same held-out forecast metric used in the main inventory Q2 pipeline.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Small MLP	Gradient-boosted trees	0.60	0.043	4/10
0.25	Small MLP	Gradient-boosted trees	0.60	0.059	4/10
0.50	Small MLP	Gradient-boosted trees	0.50	0.128	5/10
1.00	Small MLP	Moving average (7)	0.10	0.684	9/10

Table A17: Expanded-family inventory Q2 robustness summary for the broader comparison set.

In this broader comparison, agreement is mixed at 0.00 and 0.25, falls further by 0.50 and 1.00, deployed-suboptimal counts rise from 4/10 to 9/10, and moderate/high-friction mismatch strengthens. The mixed zero/low-friction rows do not overturn the narrower main claim from the five-family inventory comparison.

I Why Forecast Ordering Need Not Be Preserved

Proposition. A proper forecast-side score need not be order-preserving for deployed utility under a fixed frictional action interface.

Reversal condition. If a forecaster’s forecast-side score advantage is smaller than the extra friction-induced utility loss from its induced action path, deployed ordering can reverse.

Proof sketch. Properness concerns truthful prediction scoring. Deployed model selection uses the composed object forecast $\rightarrow$ action proposal $\rightarrow$ interface $\rightarrow$ realized utility. Therefore forecast-side ordering need not be preserved after action mediation.

Family	Input	Stage 1	Stage 2	Seeds	Representative rule
Naive persistence	fixed heuristic	–	–	final 10	fixed ex ante
Moving average (7)	fixed heuristic	–	–	final 10	fixed ex ante
Linear AR	legacy tabular	5	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Small MLP	legacy tabular	18	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Small GRU	sequence (lookback 28)	6	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Regularized linear + lag search	lagged tabular	36	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Gradient-boosted trees	lagged tabular	216	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Large MLP	legacy tabular	108	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
GRU variant	sequence	81	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config

Table A15: Standardized tuning protocol for the stronger-baseline comparison. All tunable families share the same split, validation criterion, and two-stage selection structure; the two simple heuristics remain fixed ex ante.

Family	Representative	Validation mean	Validation median
Naive persistence	fixed ex ante	-2.911	-2.981
Moving average (7)	fixed ex ante	-3.265	-3.251
Linear AR	alpha=0.0001	-2.449	-2.512
Small MLP	lr=0.001, wd=0.0, batch=32	-2.331	-2.390
Small GRU	lr=0.001, batch=32	-2.628	-2.571
Regularized linear + lag search	lag=7, lasso, alpha=0.1	-2.418	-2.410
Gradient-boosted trees	lag=14, trees=100, depth=5, lr=0.1, leaf=10	-2.425	-2.279
Large MLP	w=256, d=3, drop=0.0, lr=0.001, wd=0.0001	-1.954	-2.012
GRU variant	L=28, h=128, layers=2, drop=0.0, lr=0.001	-2.602	-2.599

Table A16: Validation-selected representatives for the broader inventory Q2 family sweep. Family representatives are chosen by validation negative MAE only.